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Quick Sort

O(n logn) time , O(logn) space

l								h
6	3	7	4	9	2	1	5	8

2	3	7	4	9	6	1	5	8

l				p-1	p	p+1		h
2	3	5	4	1	6	9	7	8
					i			
					j			

3	2	5	4	1

1	2	5	4	3

Steps

1) choose pivot element
2) put it in right place (loop)
3) partition
4) Recursion

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Quick Sort (First Element as pivot)

l					p			h
2	3	5	4	1	6	9	7	8
					j	i		

l	p			h		l		h
1	2	5	4	3		9	7	8
	j	i				p	i	j

val <= pivot

val > pivot

Swap l with j

Steps

1) choose pivot element
2) partition
3) put it in right place
4) Recursion

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Quick Sort (Last Element as pivot)

swap

6 3 7 4 5 2 1 9 8

j i

i ke saath swap (last pivot)

Steps

1) choose pivot element
2) partition
3) put it in right place
4) Recursion

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1	2	3	4	5	6	7	8	9
p								

→ $O(n^2)$ time

Middle Element as pivot (Better)

→ (Code HW)

l		p						h
1	2	4	7	3	5	6	9	8
	j	i						

Swap with 'i'

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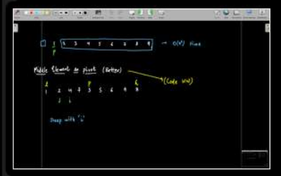
Quick Select

k^{th} smallest

M-1 Sort → nlogn return arr[k-1]

M-2 Heaps → nlog k

M-3 Quick Select → O(n) time



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Quick Select

Q// k^{th} smallest element

a b c d e f g

d b c a e f g

sort

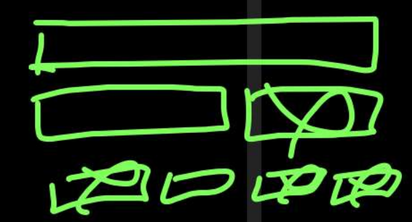
3rd smallest

Sort karne ki need nahi hai.

Search space

$n \to \frac{n}{2} \to \frac{n}{4} \ldots$

$n + \frac{n}{2} + \frac{n}{4} \ldots \approx 2n$



HW k^{th} largest element

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Quick Select

Q// k^{th} smallest element

$k = 4$

l $p-1$ p h

10 30 40 50 20 | 60 | 70 90 80

j i

5 → 6th smallest element

l

l $p+1$ h

10 | 30 40 50 20

p i

j

1st smallest

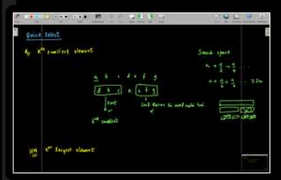
p → 3rd smallest

20 | 30 | 50 40

j i

p → 5th smallest

40 | 50

l i

j